

1.1 About EverShop E-Commerce Platform EverShop is a modern, open-source e-commerce platform built using **Node.js**, **React**, and **GraphQL**, with **PostgreSQL** as its primary relational database. It provides a modular and scalable architecture designed for containerized web applications.

1.2 Deployment & Architecture in our Environment In this project, EverShop was deployed using **Docker Compose**, orchestrating two primary containers on an isolated bridge network (myevershop):

- **app container:** Hosts the Node.js / React application serving HTTP web traffic on port 3000.
- **database container:** Hosts the PostgreSQL 16 database engine storing application state, users, and catalog data.

2.1 Network Topology Overview

The network topology was designed and deployed using **Containerlab** to simulate a multi-vendor, multi-subnet routing environment:

- **MikroTik Router (\$R_2\$):** Positioned on the left side, serving as the gateway for **Client 1 (\$C_1\$)**.
- **Nokia SR Linux Router (\$R_1\$):** Positioned in the center, acting as the main gateway for **Client 2 (\$C_2\$)** and the **Proxmox / EverShop Application Node**.
- **Inter-Router Link:** A dedicated point-to-point connection linking \$R_2\$ (MikroTik) and \$R_1\$ (Nokia) to enable inter-subnet traffic routing.

4.1 Nested Virtualization Limitations

During the deployment of the guest VM inside Proxmox VE (which itself was hosted on VirtualBox), CPU execution stalls were observed during boot up.

Technical Explanation: This behavior is caused by hardware virtualization instruction translation overhead passing through multiple hypervisor layers (Windows Host VirtualBox Proxmox VE Guest VM), resulting in KVM kernel stalls.

Verification: The validity of the Docker stack, network bridge, and YAML topology configurations was successfully confirmed and validated independently on the bare-metal host level.

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1

General

Owner *

 DuhaYameen

Repository name *

EverShop

✓ EverShop is available.

Great repository names are short and memorable. How about [fantastic-barnacle?](#)

Description

0 / 350 characters

2

Configuration

Choose visibility *

Choose who can see and commit to this repository

Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off ☐

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

Create repository